Smart Flow System Using Soil Moisture Sensors and IoT

I. Introduction

As Earth's surface is covered 71 percent of water ,In most region agriculture is performed manually or time-based as this leads to under-irrigation or over-irrigation.Smart Madai are used in gardens, greenhouses and agriculture field to supply water and suitable farming practices.As other solution are expensive and unsuitable for farmers.Instead of watering crops manually ,the system continuously monitors the moisture which waters the crop only when it is essential and makes the farmers to easier their work.It also helps the farmers to avoid common problems like too much of water used or less water which can damage crops .As we designed this for both drip and surface irrigation.When crop resives the correct amount of water at a right time ,the root grows stronger and healthier.This reduces water wastage and makes the crop to grow .This smart Madai is designed, this works on the basis of soil moisture sensor of real time weather condition datas.As soil moisture sensor is used to check the soil is dry or wet.This system automates the supply of water based on the Environmental Conditions(Soil Moisture and Weather).In The Microcontroller collects data of soil moisture sensors,temperature and humidity.In this system of solution is cost-effective, scalable for small and medium scale farmers.This creates the dataset on the basis Of Soil Moisture level,RainFall(mm),Irrigation Duration(Sec).If the soil moisture level falls below the required level system automatically activates the water to flow in each channels to irrigate the field.When the required moisture level is reached ,the pump is turned off.This system is also useful in dry season when more water is required.As we implemented the process to work through our mobile website for all farmer users .As this makes the agriculture easier, smarter and more efficient .According To the study of energy Nexus [1]Water wastage and inefficient water use in agriculture.Does not include practical fielding.By referring another study ff Akubattin.V..et al [2] Farmers must manually switch water pumps ON and OFF which leads to waste of time and water

II. Literature Survey

Smart irrigation systems have become an important research area due to the increasing need for efficient water management in agriculture. Traditional irrigation methods often lead to water wastage and inefficient crop management, which has driven the development of automated and intelligent irrigation systems.

Gurao et al. [1] proposed an IoT-based smart irrigation system that uses sensors to monitor soil conditions and automate water supply, improving water efficiency. Similarly, Rawal [2] developed a system using soil moisture sensors and IoT technology to control irrigation, demonstrating reduced manual effort and better crop management.

A review by ScienceDirect [3] highlights the importance of IoT in modern irrigation, explaining how real-time data collection and automation improve irrigation efficiency. Dharashive et al. [4] further enhanced this concept by integrating artificial intelligence with IoT, enabling predictive and intelligent irrigation decisions.

Machine learning-based irrigation systems have also been explored, where a study in Computers and Electronics in Agriculture [5] demonstrated how predictive models can optimize water usage based on environmental data. Bhandari et al. [6] implemented a practical IoT-based irrigation system using sensors and microcontrollers, showing effective real-time monitoring and automation.

An IEEE conference paper [7] discussed the implementation of smart irrigation using IoT, focusing on system architecture and communication techniques. Sensor-based irrigation systems without IoT integration were also studied [8], showing that automation can still improve efficiency even in basic setups.

Further studies [9], [10] focused on system design and implementation, emphasizing reliability and scalability in agricultural applications. Wireless sensor network-based irrigation systems [11] have also been proposed to enhance communication and data collection over large agricultural fields.

Real-time monitoring systems using IoT [12] provide continuous data updates and remote control capabilities. Precision agriculture techniques [13] integrate IoT-based irrigation to optimize crop yield and resource usage. Low-cost implementations using ESP32 [14] make these systems affordable and accessible for farmers.

Finally, cloud-based irrigation systems [15] enable data storage, remote access, and advanced analytics, improving decision-making and overall system performance.

III. Methodology

A. System Overview

The smart irrigation system is an automated solution designed to supply water to crops efficiently without manual work.It uses sensor to monitor the condition of the soil and ensures the plants receive the right amount of at the right time.Instead of manually operating the madai(water channel),The system controls flow automatically,helping to reduce water wastage and improve farming efficiency. In this system,a soil moisture sensor is placed in the field to continuously

TABLE I
LITERATURE SURVEY OF SMART IRRIGATION

Ref	Existing Work Focus	Method Used
[1]	Basic IoT Irrigation	Soil moisture sensor + automation
[2]	Sensor-based irrigation	Threshold-based control
[3]	IoT review systems	Data collection
[4]	AI + IoT irrigation	Predictive models
[5]	ML-based irrigation	Data prediction
[6]	Practical IoT system	Sensor + microcontroller
[7]	IoT architecture	Communication focus
[8]	Sensor automation	Basic irrigation
[9]	Agriculture IoT	General monitoring
[10]	System design	Hardware implementation
[11]	Wireless sensor network	Data transmission
[12]	Real-time monitoring	Continuous data
[13]	Precision agriculture	Optimization
[14]	ESP32 irrigation	Basic automation
[15]	Cloud-based system	Data storage

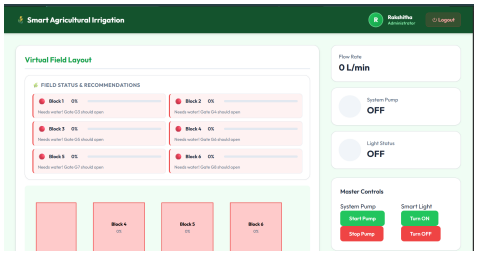

Fig. 1. dashboard1

check whether the soil is dry or wet.The data collected from the sensor is sent to the microcontroller such as Arduino or ESP32,which acts as the brain of the system.Based on a predefined threshold value,the controller decides whether irrigation is needed.If the soil moisture level is low,the system turns on the water pump or opens the solenoid valve to allow water to flow through the madai. Once the soil reaches the required moisture level,the system automatically stops the water supply by turning off pump or closing the valve.This continuous monitoring and automatic control help in maintaining optimal soil conditions for crop growth.Overall, the smart irrigation system saves water,reduces human effort,and increasing agricultural productivity by providing a simple and efficient irrigation method.

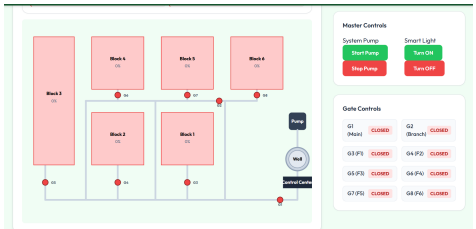

Fig. 2. dashboard2

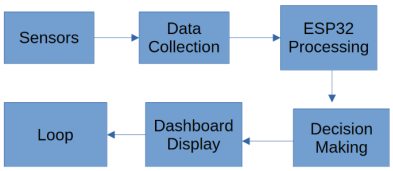

Fig. 3. Collection of datas

B. Data Collection

Data collection is an important part of the smart irrigation system,where information about the soil and environment is gathered continuously using sensors.The main data collected is soil moisture,which indicates whether the soil is dry or wet.A soil moisture sensor is placed in the field,and it measures the water content in the soil at regular intervals.Along with this,other sensors like flow sensors can used to collect data that affects plant growth. The collected data is sent to the microcontroller,where it is processed for decision-making.The system collects data repeatedly,for example,every few seconds or minutes,to ensure accurate monitoring.A water level sensor may also be used to check the availability of water in tank,preventing the pump running from when there is no water. The continuous data collection helps the system understand the real-time condition of the field and make proper irrigation decision.By using accurate sensor data,the system avoids overwatering and underwatering,ensuring efficient use of water and better crop health.

C. System Integration

The smart flow system integrates multiple hardware and software components to work as a single automated unit for efficient irrigation management.The power supply unit(Li-Ion battery and DC-DC converter) provides regulated voltage to all components including the ESP32,sensors and relay module.The ESP32 microcontroller acts as the central controller, connecting and coordinating all parts of the system. Various sensors such as soil moisture ,water flow and pressure sensors are interfaced with the ESP32 to continuously collect real-time environmental and system data.These sensors send signals to the ESP32,Where the data is processed and analyzed. Based on predefined conditions ,the ESP32 send control signal to the relay module,which switches the DC pump ON or OFF.The solenoid valves and motor are also controlled to regulate water distribution across different irrigation channels.

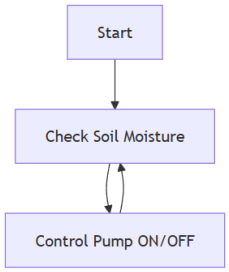

Fig. 4. Data flow

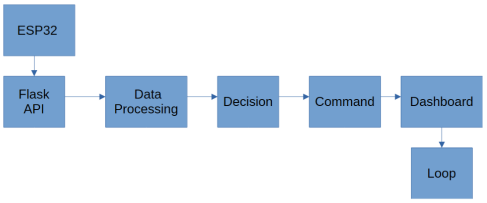

Fig. 5. Backend process

Fig. 6. Frontend process

Simultaneously,the ESP32 uses its built-in WiFi module to transmit sensor data to a web dashboard,enabling real-time monitoring of field conditions such as moisture level and water flow status.The web interface also allows users to manually control the irrigation system,overriding automatic operation when needed. The water flow sensor and pressure sensor provide feedback to ensure that water is properly reaching the field.If any abnormal condition such as no flow,leakage or blockage is detected,the system generates an alert notification on the dashboard. All components work together in a continuous loop,where data is collected,processed,transmitted and used for decision-making ensuring the integration of sensing,control,communication and user interaction for an efficient and reliable smart flow system.

D. Backend Implementation using Flask

The backend of the smart irrigation system is developed using the Flask framework, which is a lightweight web framework written in python.It acts as a bridge between the hardware and the user interface.The backend is responsible for receiving sensor data,processing it,storing it,and sending control commands to the irrigation system. In this implementation,the microcontroller sends sensor data such as soil moisture,temperature,and water level to the flask server through APIs.Flask handles these requests using defined routes,where the incoming data is processed and stored in a database.The backend also applies decision logic,such as checking whether the soil moisture is below the threshold,and decides whether to turn the irrigation system ON or OFF. Flask also provides APIs for users to monitor and control the system remotely.A user can view real-time data on web page or mobile app and manually control the pump if needed.The backend sends commands back to the microcontroller to operate the relay and control the water flow.Overall,Flask makes the system smart and connected by enabling communication,automation,and remote access in the irrigation system.

E. Frontend Development using HTML & CSS

The frontend of the smart irrigation system is developed using HTML and CSS to create a simple and user-friendly interface.It allows users to view system data and control irrigation easily through the web page.The frontend acts as a visual part of the system where all the information like soil moisture and water level is displayed clearly. Using HTML, the basic structure of the web page is created,such as heading,buttons,tables and labels to show the sensors reading.CSS is used to style the webpage by adding colors,layouts and proper alignments,making the interface attractive and easy to understand.For example,different colors can be used to indicate soil condition and buttons can be provided to manually turn the pump ON or OFF. The frontend communicates with the flask backend through APIs.It fetches real-time data from the server and displays it on the screen.Users can also send commands through buttons,which are passed to the backend and then executed by the hardware.Overall,the HTML and CSS frontend provides a simple and effective way to monitor and control the smart irrigation system remotely.

F. Power Management

Power management in the Smart Irrigation System is designed to ensure efficient and reliable operation of all components while minimizing energy consumption. The system is powered using a 3.7V Li-Ion battery pack, which is connected to a DC-DC variable converter to regulate and step up/down the voltage to required levels such as 3.3V, 5V, and 12V for different components. The ESP32 microcontroller operates at 3.3V, while sensors and relay modules typically require 5V, and components like the DC pump and solenoid valves may require 12V. The DC-DC converter ensures stable voltage supply to each component, preventing damage and ensuring proper functioning.

To improve efficiency, the system uses power optimization techniques such as turning ON high-power components (pump, valves) only when required. The ESP32 can also operate in low-power or sleep mode when the system is idle, reducing battery consumption. Additionally, the system monitors battery levels (optional) to prevent deep discharge and ensure long-term operation. Efficient wiring and proper grounding are maintained to avoid power losses. Overall, this power management strategy ensures stable operation, energy efficiency,

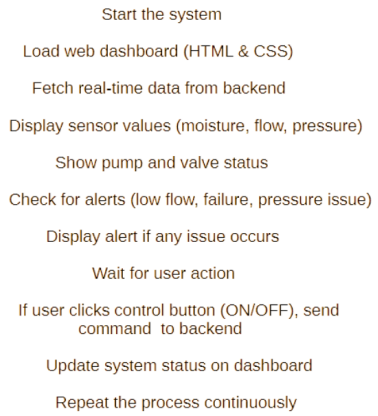

Fig. 7. Users data flow

longer battery life, and reliable performance of the smart irrigation system.

G. User Interface Design

The User Interface (UI) of the Smart flow System is designed as a web-based dashboard that provides real-time monitoring and control of the irrigation system in a simple, clear, and user-friendly manner. The interface is developed using HTML and CSS, making it accessible from devices such as phones, tablets, and computers. The dashboard displays live sensor data, including soil moisture, water flow rate, and pressure levels. These values are presented using visual elements such as graphs, gauges, and status indicators, allowing users to easily understand field conditions. The pump status (ON/OFF) and valve conditions are also clearly shown. The UI includes control features that allow users to manually operate the system, such as turning the pump ON or OFF and controlling solenoid valves for different irrigation channels. This provides flexibility alongside the automatic operation of the system. Additionally, the interface supports an alert notification system, which informs users about issues like low water flow, pump failure, or abnormal pressure conditions. These alerts are displayed prominently to ensure quick response. The UI is designed to be responsive and easy to navigate, ensuring that even users with minimal technical knowledge can efficiently monitor and manage the irrigation system remotely.

H. Architecture diagram

I. Model Training and Validation

In the Smart flow System, model training and validation can be applied if data-driven or predictive techniques are used to optimize irrigation. During the training phase, historical data such as soil moisture, and irrigation patterns are collected and used to develop a model that can predict when irrigation is required. This model learns patterns in environmental conditions and water requirements of crops. In the validation phase, the trained model is tested using new or unseen data to evaluate its accuracy and reliability in predicting irrigation needs. The

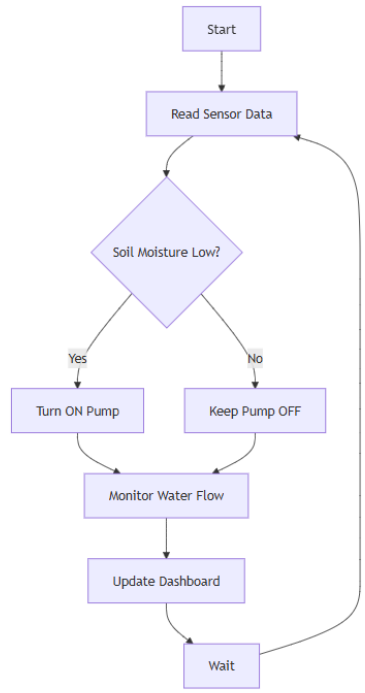

Fig. 8. Architecture Diagram of Flow

Battery	Converter	Power Distribution	Operation	Control Usage	Monitor Battery	Loop

Fig. 9. Datas of Power Flow

system checks whether the model correctly decides when to turn the pump ON or OFF based on real conditions. If the predictions match actual requirements, the model is considered valid. This process improves the system by enabling intelligent and predictive irrigation, reducing water wastage and enhancing efficiency compared to simple threshold-based methods.

IV. Results and Discussion

The Smart Irrigation System was successfully designed, implemented, and tested to provide an automated and efficient irrigation solution using IoT technology. The system effectively integrated hardware components such as the ESP32 microcontroller, soil moisture sensor, temperature and humidity sensor, water flow sensor, relay module, pump, and solenoid valves with a web-based monitoring dashboard. During testing, the sensors were able to continuously collect real-time data from the field. The soil moisture sensor accurately detected whether the soil was dry or wet, and based on this data, the ESP32 automatically controlled the water pump. When the moisture level dropped below the predefined threshold, the system activated the pump, and when sufficient moisture was reached, the pump was turned OFF. This ensured proper irrigation without overwatering. The water flow sensor successfully

monitored the flow of water in the pipeline. In cases where water flow was not detected despite the pump being ON, the system generated alert notifications, indicating possible issues such as pipe blockage or pump failure. The pressure sensor (if used) further enhanced monitoring by detecting abnormal pressure conditions. The web-based dashboard provided a clear and user-friendly interface for displaying real-time sensor data such as soil moisture and water flow. Users were able to monitor system status remotely and manually control the pump and valves when needed. The communication between the ESP32 and the backend server was reliable, ensuring smooth data transfer and control operations. The results demonstrate that the system significantly reduces water wastage by supplying water only when required. It also reduces manual labor and improves irrigation efficiency, making it suitable for modern agricultural practices. However, the performance of the system can be affected by factors such as sensor calibration errors, unstable internet connectivity, and environmental variations. Proper calibration and stable network conditions are essential for optimal performance. Overall, the Smart Irrigation System proved to be a reliable, cost-effective, and scalable solution for automated irrigation, contributing to efficient water management and improved crop productivity.

TABLE II
PROJECT RESULTS

Feature	Outcome
Automation	Successful
Monitoring	Real-time
Water Saving	High
Efficiency	Improved

V. CONCLUSION

The smart flow system is an efficient and intelligent solution for modern agriculture that uses sensors,automation, and IoT technology to manage water usage effectively.By continuously monitoring soil moisture and environmental conditions,the system ensures the irrigation is performed only when necessary,reducing water wastage and improving crop health.The integration of ESP32,sensors,a web-based dashboard,and alert mechanism provides real-time monitoring,automatic control,and remote accessibility.Overall,the system enhances irrigation efficiency,minimizes manual effort,and supports sustainable water management,making it a practical and scalable solution for smart farming.

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